



DEALER BEST PRACTICE SERIES

Managing Contract Health
Utilizing Metrics

Application Maintenance Site Management Component Rebuild Safety **MARC Management**

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1.0 Introduction

The two critical success factors to contract profitability are the Project Manager and effective measurement of contract performance. The Project Manager is the primary decision maker and represents the dealer's best interests on the mine site. Measuring contract performance will be the gauge of contract profitability but will also provide a process to indicate how well the project manager is making decisions. The ultimate objective in measuring contract performance is to provide early warnings of potential contract shortfalls to the dealer management group. If an operation is projecting an unfavorable trend, Dealers and Caterpillar can partner together to apply resources to turn the trend around before it is too late. Time is the key advantage to turning around a negative trend. The earlier the measuring process is initiated, the earlier dealer management will realize a project needs help. The best-case scenario initiates the measurement process as soon as the project becomes operational. The Preventive Maintenance service routine can be measured against standard job variance or a time requirement guide right from the very first PM service performed. This practice can set the standard for the entire project and start from day one.

2.0 Best Practice Description

Using 6 Sigma and input from dealers, Cat Global Mining determined the top five metrics to measure contract profitability in conjunction with dealer support. They are: **revenue/expense variance, contractual availability, payload compliance, component management, and budget/cost variance**. These metrics are most important in providing the "early warnings" required to prevent a potentially unprofitable contract.

Two time frames are considered for measurement: "to-date" and "projected". The "to-date" time frame indicates how the project is currently performing with the existing management, resources, and operating principals. The "projected" time frame looks into the future and projects how well the project will be performing at the end of the contract. Looking at the five metrics from these two points of view will deliver a clear indication of contract profitability.

REVENUE/ EXPENSE VARIANCE:
billings – costs

CONTRACTUAL AVAILABILITY:
$$\frac{\text{actual uptime}}{\text{contracted uptime}} = \%$$

PAYLOAD COMPLIANCE:
% of loads > 110% of target
% of loads > 120% of target
mean average payload/month

BUDGET/COST VARIANCE:

operation actual
budget – cost

Revenue/ Expense Variance: This variance is the difference between the currency collected from a customer through billings based on the service meter readings and the costs applied to deliver those obligated services.

Contractual Availability: Contractual Availability is defined as the ratio of time a machine is capable of functioning in the intended operation (available hours) to the total calendar hours in the period under consideration, expressed as a percentage.

Availability (Fleet Level)

Last 6 months	Jan	Feb	March	April	May	June
Contract Availability - Actual	87.00%	89.00%	89.00%	85.00%	87.00%	85.00%
Guarantee Target	88%	88%	88%	88%	88%	88%

Fleet Availability		
3 month rolling	6 month rolling	12 month rolling
88.00%	88.90%	89.00%

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Managing Contract Health Utilizing Metrics

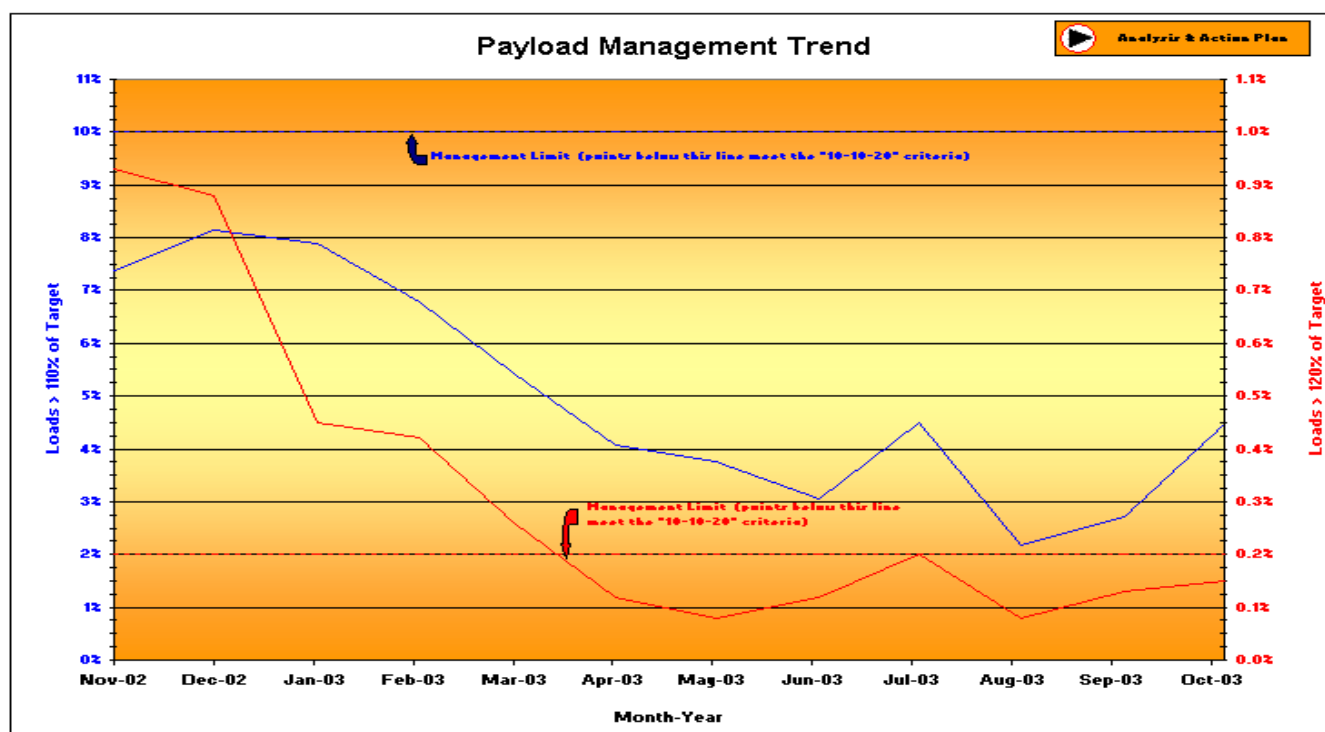
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The calculation of available hours is not a pure calculation since the result is amended by downtime hours that are specifically excluded or limited by the terms of the contracts. If there is an availability obligation with a penalty, the penalty paid to the customer for non-performance will impact contract profitability. The ideal time-period for contractual availability is a rolling 12 month average.

Payload Compliance: Adhering to the 10/10/20 payload policy is important in maintaining high operational performance for trucks and loaders. Monitoring and reporting payload a dealer can partner with a customer to maximize return on equipment investment. The elements tracked in payload compliance are: % of loads greater than 110% of target, % of loads greater than 120% of target, and mean average payload carried per month.



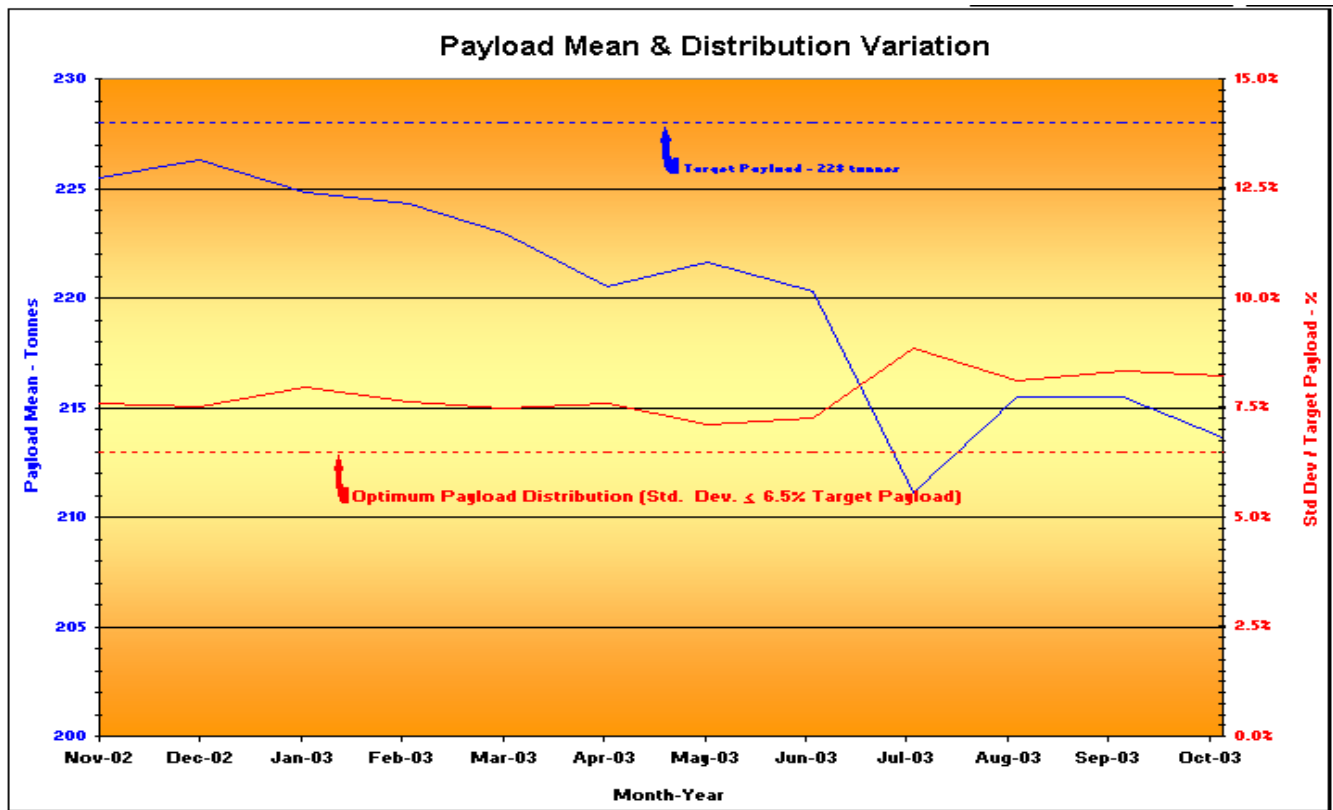
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Component Management: A thorough understanding of the site conditions and component life will be a catalyst to continuous process improvement. Understanding component life (actual life being realized vs. budgeted life) and reasons for each component removal are extremely important. The number of components required for a year must also be determined through proper tracking and planning. A project manager will directly impact contract profitability by maximizing the life a component will deliver.

Major Component Planning (Fleet Level)

	Component Changeout History Current Year-To-Date							Component Performance Over Contract Life-To-Date					
	Budgeted Components for Current Year	Q1	Q2	Q3	Q4	Components YTD	Component YTD Variance	Budget	Min	Average	Max	Components To Date	Components Over Contract Life
Engine	0	0	0	1	0	1	-1	16000	6,855	6,855	6,855	1	25
Transmission	0	0	0	1	0	1	-1	12000	8,063	8,063	8,063	1	40
Torque	0	0	0	2	0	2	-2	16000	4,354	6,645	8,336	2	26
Differential	0	1	1	3	0	5	-5	15000	5,230	7,023	8,617	5	32
Final Drives_Left	0	1	0	1	0	2	-2	15000	3,622	6,120	8,617	2	32
Final Drives_Right	0	0	0	2	0	2	-2	15000	11	5,863	8,360	3	32

	Removal Tracking						YTD Total
	PCR	A	B	C	D	E	
Engine	6						
Transmission	0						
Torque	0						
Differential	3						
Final Drives_Left	4						
Final Drives_Right	4						

Reason for Removal Key	
PCR: If a component achieves 30-32% of planned budget	
A)	Convenience / Synchronized to other changeout events
B)	Required Update (PIP / PSP)
C)	Cease to function / Failed
D)	Dealer Redo (within x hours of repair)
E)	Failed within Warranty Period
F)	Condition Monitoring

This planning spreadsheet shows both short-term and long-term component consumption. The Year-To-Date area tracks components, by type, as they are used. The Contract Life-To-Date area compares actual component life against budgeted hours, and Removal Tracking specifies the reason for each component removal.

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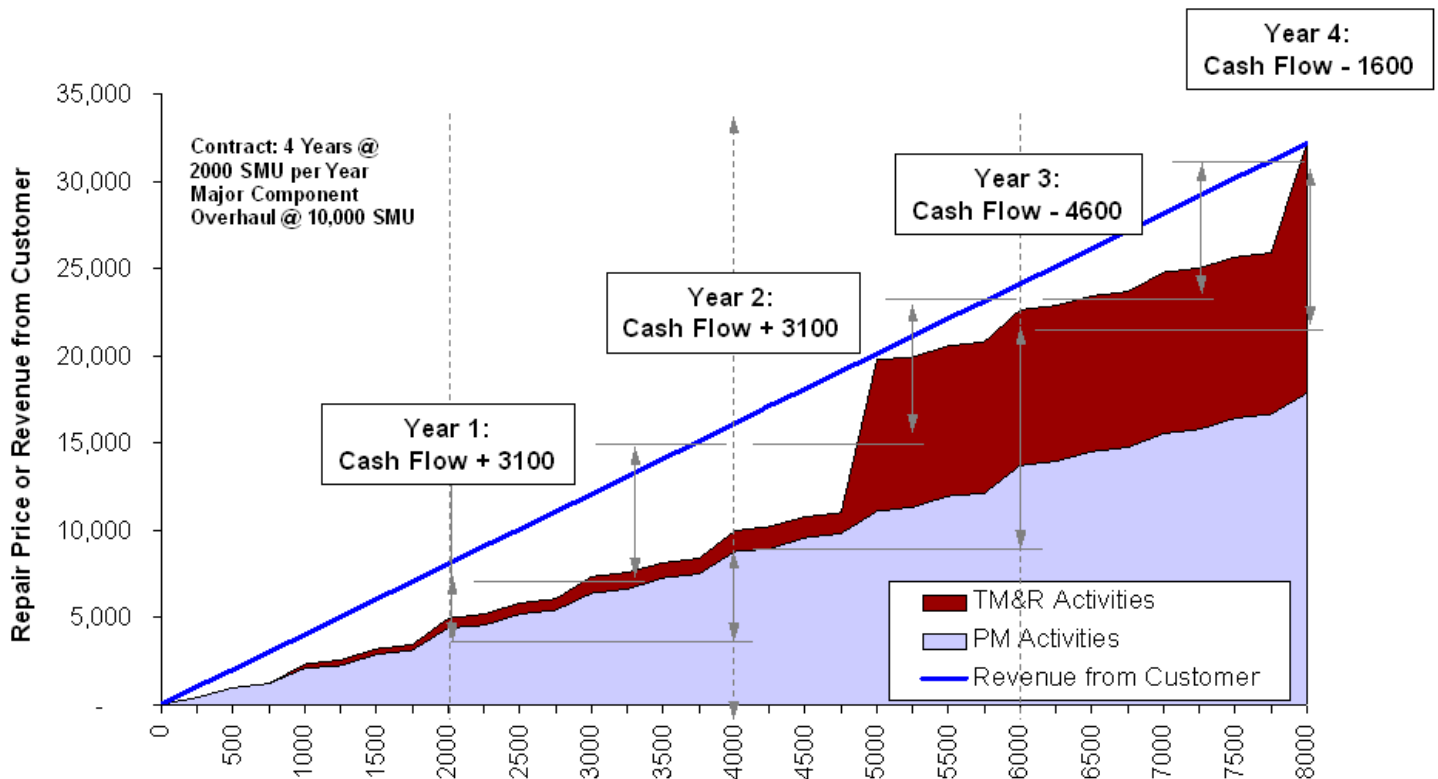
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Budget/Cost Variance: This variance is the difference between the billing rate per hour charged to the customer and the actual budget created to deliver the parts and service as agreed in the contractual obligation.

The illustration below shows the relationship between revenue (blue line) and expenses (solid colors, PM's & Repairs) over an 8,000 hr term. If the contract is operating as intended, the expenses should never be greater than revenue on a cumulative basis. If expenses do achieve a higher level than revenue, further evaluation will be required to determine the root cause.



Budget/cost variance is simply the difference between the dollars spent for a specific budget item and the target cost.

Ex: 1000 Hr PM: Budget = \$1200, Cost = \$1050, $\Rightarrow$ Variance = \$150

Year / Quarter	SMU (Avg/Unit)	% Complet	To Date Actual			Projected			Projected Variance Breakdown			Adjusted GM - %	Adjusted GM - \$\$\$'s
			Revenue	Expense @ Sell	Variance	Revenue	Expense @ Sell	Variance	Cost	Escalation	Budget		
Q3_2005	8,578	14.3%	\$3,574,752	\$318,885	\$2,655,867	\$19,456,041	\$21,206,037	(\$1,749,996)	\$435,434	(\$885,430)	(\$1,353,340)	24.5%	\$5,186,585

The above table includes terms introduced from the ANALYZER™ software tool: projected cost variance, escalation variance, and budget variance.

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Cost Analysis @ Sell (Fleet Level)

SORT: Actual Sales & Expenses Life To Date				
Component Description	Prioritized By % of Actual	Current Budget	Current Actual Sales	Actual Sales Variance
1000 Engine	177%	1,488,751	2,629,169	(1,140,418)
1000.UNS Engine	588%	193,274	1,136,757	(343,483)
1250 Fuelsystem	470%	283	1,331	(1,048)
1353 Radiator	239%	103,420	247,068	(143,648)
1355 Thermostat/Water	355%	1,298	4,602	(3,304)
1359 Fan Drive	119%	11,448	13,602	(2,154)
1405 Alternator	196%	6,544	12,802	(6,258)
1451 Air Starting Motor	102%	42,345	43,720	(1,375)
1802 Refrigerant Comp	179%	13,073	23,346	(10,273)
3000.UNS Power Train	127%	183,163	241,170	(58,007)
3101 Torque Converter	103%	145,807	150,441	(4,634)
3253 Drive Shaft	89%	62,362	55,743	6,619
3258 Differential	92%	345,687	319,747	25,940
4000.UNS Drive Train		0	446,172	(446,172)
4001.LR Final Drive & B	109%	1,035,891	1,132,015	(96,124)
4001.RR Final Drive & B	107%	1,035,891	1,110,791	(74,900)
5000.UNS Hydraulics	91%	105,474	96,079	9,395
5050 Hydraulic System		0	2,589	(2,589)
7000.UNS Machine		0	335,095	(335,095)
7500.UNS Servicing		0	21,046	(21,046)
9000.UNS Miscellaneous		0	-34,155	34,155

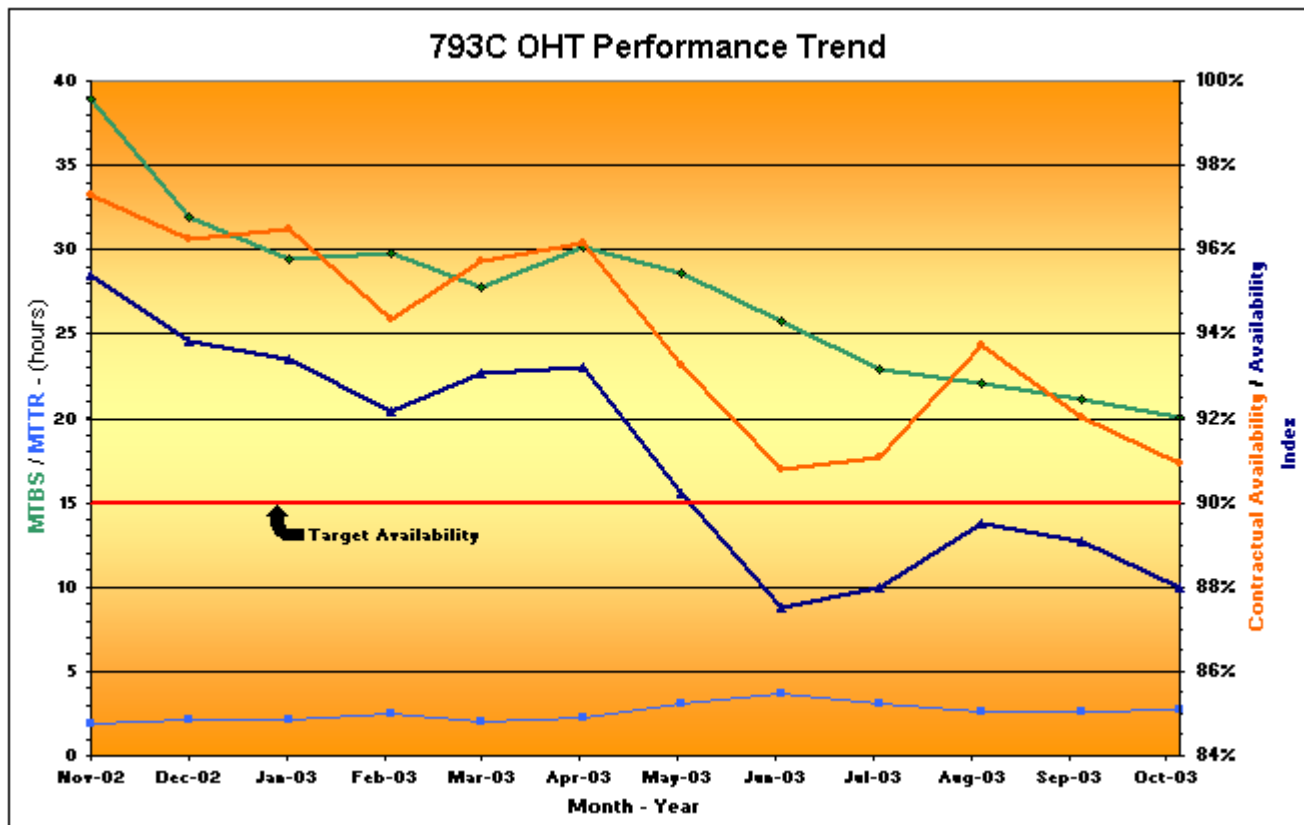
SORTED By: Projected To Contract End				
Component Description	Prioritized By % of Actual	Project Budget	Projected Sale	Projected Sales Variance
4000.UNS Drive Train		0	446,172	(446,172)
7000.UNS Machine		0	335,095	(335,095)
7500.UNS Servicing		0	21,046	(21,046)
5050 Hydraulic System		0	2,589	(2,589)
9000.UNS Miscellaneous		0	-34,155	34,155
1000.UNS Engine	474%	253,022	1,188,326	(345,304)
1250 Fuelsystem	378%	379	1,431	(1,052)
1355 Thermostat/Water Tem	260%	2,006	5,222	(3,216)
1353 Radiator	157%	183,024	295,384	(106,360)
1405 Alternator	154%	10,013	15,404	(5,391)
1802 Refrigerant Compress	134%	19,354	26,787	(6,833)
3000.UNS Power Train	122%	247,644	301,076	(53,432)
1000 Engine	116%	2,715,771	3,155,034	(439,263)
1359 Fan Drive	112%	17,714	19,868	(2,154)
4001.LR Final Drive & Brake	111%	1,483,259	1,643,292	(160,033)
4001.RR Final Drive & Brake	109%	1,483,259	1,622,068	(138,809)
3101 Torque Converter	102%	265,276	269,910	(4,634)
5000.UNS Hydraulics	94%	137,395	129,790	7,605
3253 Drive Shaft	88%	92,383	81,762	10,621
1451 Air Starting Motor	86%	70,153	60,045	10,108
3258 Differential	82%	527,365	433,295	94,070

Additional elements tracked in this report:

MTBS: Mean Time Between Stops

MTTR: Mean Time To Repair

The following table illustrates the format for the data collection process that will generate the 793C OHT Performance Trend. The supplemental document [Performance metric.pdf](#) contains additional information on MTBS and MTTR.



Additional metrics:

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Projected Cost Variance (PCV): The contract budget created to build the rate per hour is used as a baseline. As expenses are applied to the contract through the work order system, these expenses are continually compared to the original baseline. The PCV is an indicator of financial performance compared to the original budget from a total contract perspective. If this indicator is negative, further analysis will be required to understand the source of the trend and action plans put into place to resolve.

Escalation Variance: Escalation variance is defined as the relationship between parts price increases to cost of goods sold and the ability to raise the billing rate to the customer. Analyzer has the ability to calculate this ratio and display in the dollar format.

3.0 Implementation Steps

Contract Health implementation will require the cooperation of several functional areas within the dealership, especially without the use of ANALZYER or AMT. These software tools are capable of creating projections through contract end of revenue and contract expenses directly from input financial information or reports. If dealers do not use projection software, the department responsible for customer billings will provide the “to-date” and “anticipated” contract revenue values. The dealership department responsible for tracking contract cost will provide the to-date expenses and the anticipated expenses through the end of the contracts. A dealer operational area will provide contractual availability and payload compliance data.

Contractual Availability is defined as the ratio of time a machine is capable of functioning in the intended operation (available hours) to the total calendar hours in the period under consideration, expressed as a percentage. The calculation of available hours is not a pure calculation since the result is amended by downtime hours that are specifically excluded or limited by the terms of the contracts. Contractual availability data will be created in the operational area responsible for tracking and reconciling downtime directly with customers. The tracking elements are: date/time down, reason for unit being declared down, length of down period, reason for down term (i.e. waiting on parts, service technicians, waiting on a service bay), the party accountable for the reason (customer, dealer, or product), and date/time returned to service. This information ideally is reconciled on a daily or weekly basis. For further detail, please refer to the attached document [Performance metric.pdf](#). This document provides specific detail for implementing a delay code management process, which will facilitate prioritizing the reasons for down time with a generic Pareto analysis.

The following table shows the data format required to establish the contractual availability trend. To display the date value as a number, refer to the attached file [Date Format.doc](#).

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Managing Contract Health Utilizing Metrics	DATE 2/20/2015	CHG NO 02	NUMBER 0107-3.3-1056

DATE	DATE VALUE	MTBS	MTTR	Avail Index	Cont. Avail	Target Avail
					$=F52/(F52+G52)$	
15-Nov-02	37575	38.94	1.88	95.40%	97.32%	90.00%
15-Dec-02	37605	31.89	2.09	93.84%	96.23%	90.00%
15-Jan-03	37636	29.44	2.08	93.39%	96.48%	90.00%
15-Feb-03	37667	29.85	2.53	92.18%	94.33%	90.00%
15-Mar-03	37695	27.78	2.08	93.05%	95.74%	90.00%
15-Apr-03	37726	30.15	2.20	93.19%	96.15%	90.00%
15-May-03	37756	28.59	3.09	90.24%	93.27%	90.00%
15-Jun-03	37787	25.74	3.68	87.50%	90.79%	90.00%
15-Jul-03	37817	22.90	3.13	87.99%	91.09%	90.00%
15-Aug-03	37848	22.07	2.59	89.49%	93.74%	90.00%
15-Sep-03	37879	21.18	2.60	89.07%	92.03%	90.00%
15-Oct-03	37909	20.04	2.73	88.01%	90.94%	90.00%

Implementing payload management will require support from the operational department. Payload management is defined as an analysis of payload distribution for a fleet of off highway trucks expressed in terms of Caterpillar 10/10/20 truck overload policy. The sources of data for measuring payload start with the gross machine weights. These can be obtained from the Caterpillar performance handbook, machine specifications sheets and information contained in the factory documentation for the 10/10/20 payload policy. Ideally, the data is obtained once the machine has been assembled and commissioned using an actual scale truck study. The actual gross machine weight is an important element for determining the actual payload the machine may haul while staying within the payload policy. Factors such as body design, tires, optional equipment, carry-back, or other debris accumulation can have a significant effect on the accuracy of published estimates. Actual payload data is obtained from the truck payload management system (TPMS) or VIMS™-TPMS reports. For further detail into payload management, please refer to the [Performance Metric.pdf](#) attachment. The following table illustrates the data format required to develop the reports that will illustrate contract performance and establish analysis trends.

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Managing Contract Health Utilizing Metrics	DATE 2/20/2015	CHG NO 02	NUMBER 0107-3.3-1056

DATE	DATE VALUE	110% Limit	120% Limit	% > 110%	% > 120%	Payload Mean	Target Payload	Standard Deviation	Std Dev / Target Payload	Optimum Payload Distribution
15-Sep-02	37514									
15-Oct-02	37544									
15-Nov-02	37575	10.00%	0.20%	7.37%	0.93%	225.50	228.00	17.32	7.60%	6.50%
15-Dec-02	37605	10.00%	0.20%	8.15%	0.88%	226.30	228.00	17.17	7.53%	6.50%
15-Jan-03	37636	10.00%	0.20%	7.88%	0.45%	224.80	228.00	18.13	7.95%	6.50%
15-Feb-03	37667	10.00%	0.20%	6.80%	0.42%	224.30	228.00	17.41	7.64%	6.50%
15-Mar-03	37695	10.00%	0.20%	5.42%	0.26%	223.00	228.00	17.08	7.49%	6.50%
15-Apr-03	37726	10.00%	0.20%	4.07%	0.12%	220.50	228.00	17.32	7.60%	6.50%
15-May-03	37756	10.00%	0.20%	3.75%	0.08%	221.60	228.00	16.24	7.12%	6.50%
15-Jun-03	37787	10.00%	0.20%	3.06%	0.12%	220.30	228.00	16.53	7.25%	6.50%
15-Jul-03	37817	10.00%	0.20%	4.49%	0.20%	211.10	228.00	20.19	8.86%	6.50%
15-Aug-03	37848	10.00%	0.20%	2.18%	0.08%	215.50	228.00	18.52	8.12%	6.50%
15-Sep-03	37879	10.00%	0.20%	2.71%	0.13%	215.50	228.00	18.97	8.32%	6.50%
15-Oct-03	37909	10.00%	0.20%	4.47%	0.15%	213.60	228.00	18.73	8.22%	6.50%
15-Nov-03	37940	10.00%	0.20%	5.90%	0.23%	217.10	228.00	18.86	8.27%	6.50%

Implementing component management will require cooperation between the operational, financial, planning and technical departments. Each dealer may be slightly different in functional responsibilities, however the elements to be tracked and reviewed will be the same. Each dealer will decide where the responsibility resides.

The process implementation starts with planning. Questions to answer include: how many of each major component will be required this year and when?, What is the contract budget PCR, and how will the dealer track actual life realized? How will actual life realized be improved? The operations or financial group will determine when components were removed, SMU, and the reason for removal. This group will be the closest to the root cause analysis for the component. The technical department will need to know why the components are removed in order to track and identify emerging component removal trends. If a trend is emerging, the technical department will be in the best position to monitor and consolidate the territory experience.

The trend information can be used in discussions for continuous improvement meetings with the customer or product health meetings at the dealership. This will be an integral part of understanding the root cause for achieving, or not achieving, desired PCR and the management process for improving current trends. This same process will be replicated for each major component in the contract. Implementing a "to-date" and "projected" budget cost variance analysis will require cooperation between planning, financial, and operations groups. The financial planning, or rate development group, will establish the target budgets for the site operations groups.

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Major Component Tracking System - Microsoft Internet Explorer

CATERPILLAR Major Component Tracking System

Component Life Spreadsheet

Main Store Dealer Code: **H160** Report Date: **22-AUG-2003**

Customer Name: **ALL** Sales Model: **793B**

Component Type: **ENGINE** Job Site: **ALL**

Component SMU Unit: **HOURS** Fuel Unit: **GALLONS**

Component S/N	Dealer/Custom ID No	Prime Product S/N	Cumulative SMU	Life Cycle	Install Date	Origin for Install	Component SMU	Removal Date	Reason for Removal
1000-1HL206-CH	555	1HL00206	UNKNOWN	1	01-Feb-99	NEW		03-Feb-99	FAILURE
2PK00184	102	1HL00021	18409	1	15-Jan-93	NEW	5,303	04-Oct-93	FAILURE
	119	1HL00075		4	04-Sep-95	UNKNOWN	11,106	17-Mar-97	PCR
2PK00167	103	1HL00022	19707	1	09-Jan-93	NEW	4,999	08-Sep-93	FAILURE
	501	1HL00042		2	17-Sep-93	UNKNOWN	1,512	28-Nov-93	FAILURE
	121	1HL00077		4	08-Nov-94	UNKNOWN	13,286	04-Sep-96	PCR
2PK00170	104	1HL00023	32073	1	25-Jan-93	NEW	8,919	18-May-94	FAILURE
	122	1HL00078		2	11-Jul-94	UNKNOWN	10,992	31-Dec-95	FAILURE
	263	1HL00157		3	26-Jan-96	UNKNOWN	11,670	29-Aug-97	PCR
	122	1HL00078		4	22-Oct-97	UNKNOWN	492	18-Nov-97	FAILURE
2PK00174	105	1HL00024	21043	1	08-Feb-93	NEW	11,014	24-Aug-94	FAILURE
	104	1HL00023		3	07-Mar-96	UNKNOWN	10,029	16-Jul-97	FAILURE
2PK00220	106	1HL00036	27364	1	21-Mar-93	NEW	9,483	20-Jul-94	FAILURE
	501	1HL00042		2	07-Sep-94	UNKNOWN	13,930	17-Jul-96	PCR
	103	1HL00022		3	02-Aug-96	UNKNOWN	3,943	16-Apr-97	FAILURE
2PK00221	109	1HL00039	25534	1	04-Apr-93	NEW	7,119	26-Apr-94	FAILURE
	108	1HL00038		2	02-Oct-94	UNKNOWN	11,681	19-Apr-96	PCR
	109	1HL00039		3	09-May-96	UNKNOWN	6,734	25-Apr-97	FAILURE
2PK00222	107	1HL00037	35510	1	25-Mar-93	NEW	5,671	07-Jan-94	FAILURE
	16035	1HL00073		2	28-Feb-94	UNKNOWN	10,321	11-Jul-95	FAILURE
	16036	1HL00149		3	20-Sep-95	UNKNOWN	12,050	10-Apr-97	PCR

This information is also analyzed for performance reporting, highlighting areas that can be used for a site continuous improvement or product health effort. Tracking actual component installs and removals is an essential element in understanding actual life realized. The Major Component Tracking System (MCTS) is an outstanding tool to accomplish this task for the dealership. MCTS is a web-based system making the information available to each

person in the dealership. MCTS has several reports to make tracking and managing actual component life a reasonable task for a dealer to complete.

CATERPILLAR Major Component Tracking System

Home | Help

Component Life Query

Main Store Dealer Code: **R460** Report Date: **19-AUG-2002**

Customer Name: **INTERCOR** Sales Model: **793B**

Component Type: **ENGINE** Job Site: **CERREJON**

Job Site Type: **COAL** Life Cycles Up to: **ALL**

Component S/N Range: **ALL**

Life	Average Component Life		Repair Type								Actual Sample Size
	Percent	Average Hours	Failure Repair		Normal Wearout		PCR		Unknown		
			Percent	Average Hours	Percent	Average Hours	Percent	Average Hours	Percent	Average Hours	
1	100%	14,569	11%	8,641	7%	11,952	76%	15,727	3%	14,427	26
2	100%	10,877	28%	6,572	16%	12,849	48%	14,014	8%	3,175	25
3	100%	6,024	66%	4,128	0%	0	16%	13,450	16%	6,179	12

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Implementing a “to-date” and “projected” Budget Cost Variance analysis will require support from several departments and will also benefit other departments. Support will be required (if an Analyzer or AMT tool is not available) to determine the “to-date” operational budgets. Operational budgets will establish the limits the site coordinators will be required to achieve when executing repair options and deploying resources. “To-date” actual data is used in analysis to monitor contract financial performance against each budget. Unfavorable budget performance can be reviewed for performance, training, site, and facility issues, which can be corrected to lead to contract profitability. The table below shows sample data to perform a budget cost variance analysis from a to-date perspective. A projection tool such as Analyzer or AMT can be used to develop a projection as well. Budget cost analysis trends can be used for dealer and customer continuous improvement projects as they are established.

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Managing Contract Health Utilizing Metrics	DATE	CHG NO	NUMBER
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Component Budget Repair Cost Vs Actual Repair Cost								August-2006
Job Number	Model Fleet No	Part Number	SOS and Description	Labour Hours On-site		Component Repair Cost		Notes
				Budget	Actual	Budget	Actual	
Site 1								
KM18803	793C-215	1470797X	503-engine	190.00	127.75	239,049.00	260,722.06	
KM24136	793C-227	8X0222X	503-front brake	21.00	22.50	16,975.00	19,657.98	
KM24985	793C-219	1503468	445-brake c/pump	4.00	9.00	7,816.00	2,784.32	
KM22629	793C-217	1063722X	503-rear s/cyl	14.00	15.00	8,871.00	13,280.10	
KM22628	793C-217	1063722X	503-rear s/cyl	14.00	15.50	8,871.00	13,094.91	
KM14786	D10R-303	1102656X	501-final drive	24.00	31.00	17,447.00	27,428.86	BER....reman used to repair
KM22641	854G-340	1539939X	501-lift cylinder	6.00	0.00	3,054.00	4,940.79	
KM24598	793C-214	9T8912X	503-steering cyl	14.00	11.00	5,872.00	6,343.99	
KM25820	D10R-303	4T5112	445-ripper lift cyl	6.00	41.75	4,105.00	9,196.12	
KM18648	793C-222	1590038X	503-rear wheel gp	30.00	49.25	86,768.00	97,054.07	
KM25549	793C-227	8X0222X	503-front brake	21.00	37.50	16,975.00	22,323.79	
KM25821	D10R-303	4T9977	445-ripper tilt cyl	6.00	26.00	3,548.00	10,599.64	
KM25444	793C-222	9U9822	445-transmission	20.00	33.00	34,282.00	30,806.37	
Site 2								
BH16396	789B-T145	3T0034X	501-transmission	24.00	36.00	35,971.00	37,705.22	\$6,000 service fee contributes to cost over-run
BH10255	793B-T268	1165808X	503-torque convt		56.00	-	16,392.07	Note: there are not any F/D events budgeted for this truck
BH14248	789B-T147	1062616X	501-wheel group	35.00	38.30	61,644.00	79,147.95	
BH14441	789B-T140	1062616X	501-wheel group	35.00	71.00	61,644.00	92,110.99	
BH16790	793B-T268	9T8912X	501-steering cyl	10.00	11.50	6,077.00	7,148.57	
BH16157	789B-T144	3T0034X	501-transmission	24.00	38.50	35,971.00	38,825.40	
BH17193	789B-T143	1062616X	503-wheel group	35.00	51.50	61,644.00	79,435.08	

4.0 Benefits

The benefits to implementing the Contract Health Best Practice are:

- Early indication to contract health
- Thorough understanding of contract profitability
- Resource management
- Asset management
- Measurements for the project manager performance

5.0 Resources Required

This Best Practice needs a dedicated resource accountable for understanding, tracking and managing contract profitability. Once trends and issues have been identified, the dealership, customers, and Caterpillar can partner together to establish action plans to mitigate contract risk.

Access to MCTS is free to dealers and can be obtained by contacting:

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Managing Contract Health Utilizing Metrics	DATE 2/20/2015	CHG NO 02	NUMBER 0107-3.3-1056

Shari Weber
 MCTS System Administrator
 Weber_Shari_Aat.com
 +1 309 675 8115

6.0 Supporting Attachments & References

Attachments:

Performance Metric.pdf	CGM whitepaper detailing the importance of mining performance metrics, also includes development of a Delay Code system.
Contract Health Template.xls	A sample contract health report ready for actual numbers to be input.
Date Format.doc	Instructions for switching date number formats in Excel.
Budget Cost Analysis.xls	Spreadsheet to compare budget and actual costs in repairs.

7.0 Related Best Practices

MCTS Used to Monitor & Manage Component Life

0906-3.1-1020

8.0 Acknowledgements

This Best practice was written by:

Don O'Neill
 MARC Manager
 Caterpillar Global Mining
Oneill_don_g@cat.com
 309.675.4861

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